Prodigy Anesthesia Quick Review

AICD

What types of patients typically have AICDs placed?	AICDs may be used in patients ranging from otherwise healthy young patients with supraventricular dysrhythmias such as Wolff-Parkinson-White syndrome to elderly patients with significant coronary artery disease or congestive heart failure who have survived a previous life-threatening arrhythmia such as ventricular tachycardia or ventricular fibrillation. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 1271.*
What type of anesthesia is typically utilized for the implantation of an AICD?	AICDs are typically implanted under sedation in an electrophysiology lab rather than the operating room. The procedure itself is not very painful, but ventricular fibrillation is induced during the procedure in order to test the device and this is distressing to the patient. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1268.*
How long after sensing ventricular tachycardia or ventricular fibrillation will an AICD discharge an electric shock?	An AICD usually discharges about 10-15 seconds after detecting VT or VF. Most of the time, delay is attributed to the charging of the capacitor. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1269.*
How long will the battery of an AICD last?	The battery of most AICDs is designed to deliver 120 shocks and typically lasts 3-6 years. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1268.*
Can a healthcare provider be shocked by a patient's AICD?	If in physical contact with a patient who has an AICD that is discharging, a healthcare provider can receive a mild shock that is not capable of causing fibrillation in the caregiver, but can be distressing while attempting to administer care to the patient. If a patient is experiencing 'electrical storm' from the defibrillator (numerous shocks are being discharged in a short period of time) it is recommended to place the patient on an ECG monitor and watch for periods of VT or VF that last about 4 seconds. This will give the caregiver several seconds warning before a shock is discharged to cease physical contact with the patient. *Davis, C. (2007). Electrical Storm and Implanted Defibrillators. Southern Medical Journal, 100, 547-548.*
Is the presence of an AICD of special concern in patients undergoing extracorporeal shock wave lithotripsy (ESWL)?	Pacemakers and AICDs are no longer considered contraindications to ESWL. AICD and lithotripter manufacturing companies generally state that AICDs are a contraindication to lithotripsy, but patients with AICDs have been shown to complete the procedure successfully. Older abdominally-implanted AICDs present a greater hazard than transvenous pacemakers. The AICD should be shut off prior to treatment and then reactivated immediately after the procedure. *Miller RD. Miller's Anesthesia. 8th ed. Philadelphia: Elsevier Saunders; 2015: 2237.*

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